2013-2014 – Master 2 – Macro I

Lecture notes #12 : Competitive search equilibrium

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Disclaimer

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Introduction

- The idea behind competitive search equilibrium is that search intermediaries could arise and internalize the congestion externalities and appropriability problems associated with search.
- This is similar to the issues arising in local public good provision and coasian internalization of externalities through contracts and property rights.

1. Fishing Clubs

Model

- A pond where the number of fish being caught is $F(N)$, where N is the number of fishermen.
- Each fisherman catches $F(N)/N$ fish
- The value of each fish is 1 and the opportunity cost of fishing per fisherman is c.

1. Fishing Clubs

Equilibrium without clubs and Optimum

- The equilibrium number of fishermen is such that
$$\frac{F(N)}{N} = c.$$
- That is
`Average productivity of fishing = cost of fishing`
- But social welfare is
$$W(N) = F(N) - cN.$$
- Thus the optimum is
$$F'(N) = c.$$
- That is
`Marginal productivity of fishing = cost of fishing.`
- The average/marginal distinction is the essence of a congestion externality.

1. Fishing Clubs
Subponds

- The pond is cut into K identical subponds
- Assumption about fishing technology : if n people fish in a subpond, they catch $F(Kn)/K$ fish
- Consistency check : if N fishermen are uniformly distributed over the subponds, they catch $KF(Kn)/K = F(N)$ fish
- Each subpond is owned by a firm which charges an entry fee p per fisherman.
- These firms compete for customers but earn positive rents due to their ownership of part of the ponds
- Competition brings their prices in line, but a firm can charge a higher price and remain competitive if in exchange for that it offers less congestion, i.e. more fish, to its clients (members)

1. Fishing Clubs
Subponds

- The value to a customer of a pond which charges p and has n people is
$$v(p,n) = \frac{F(Kn)}{Kn} - p - c.$$
- In equilibrium, all subponds such that $n > 0$ (called "active") must yield the same value to the customers :
$$\frac{F(Kn)}{Kn} - p - c = \bar{u}.$$
- This generate an equilibrium relationship between the density of fishermen at an active pond and its entry fee :
$$p(n) = \frac{F(Kn)}{Kn} - c - \bar{u}. \tag{1}$$

1. Fishing Clubs
Profit maximization

- Since owners maximize profits, they will choose the density that maximizes $np(n)$:

$$n = \arg\max np(n) = n\left(\frac{F(Kn)}{Kn} - c - \bar{u}\right).$$

- The FOC of firms is $F'(Kn) = c + \bar{u}$.
- At a symmetrical equilibrium, $n = N/K$ and therefore : $F'(N) = c + \bar{u}$.
- Since there are many (many) potential fishermen that can freely join clubs, in equilibrium $\bar{u} = 0$.
- Therefore, $F'(N) = c$.
- The equilibrium with competitive clubs is efficient.

1. Fishing Clubs

Efficient equilibrium

- The price schedule defined by (1) accurately reflects the congestion externality : clubs with more fishermen must charge a lower price.
- This makes the total profit of a club equal to the economic surplus it creates, which leads to an efficient outcome.
- The theory of competitive search (Moene (1995)) applies these ideas to the Mortensen-Pissarides matching model.

2. Search Clubs

Assumptions

- There are several ways of doing it. Here we assume "wage posting".
- firms and workers can join a search intermediary, i.e. they *assign* their vacancy and candidacy to one of those clubs.
- Vacancies and unemployed workers are only matched together if they belong to the same club.
- Each club specifies the wage that must be paid to the worker after the match has taken place.
- A club contains vacancies and unemployed workers in proportions θ, and specifies the wage w that will be paid after a vacancy is matched with an unemployed.
- The unemployed members of the club are charged f per unit of time until they are matched, while vacancies are charged g per unit of time.

2. Search Clubs

Values

- **Note :** We restrict here to stationary equilibrium, so that all $\dot{V}$ are 0.
- In equilibrium, the value of being unemployed V_u which is the maximum value offered to the unemployed across clubs.
- V_v is the maximum of posting a vacancy across clubs. With free entry , $V_v = 0$.
- Let $V_u(w, \theta, f)$ the value to the unemployed of joining a club with (w, θ, f). The asset valuation equation is

$$rV_u(w, \theta, f) = -f + \theta q(\theta)(V_e(w) - V_u(w, \theta, f)),$$

where $V_e(w)$ is the value of a job to the worker paying a wage w, which satisfies

$$rV_e(w) = w + s(V_u - V_e(w)).$$

2. Search Clubs

Equilibrium

- In equilibrium, any club who is able to attract workers must be such that $V_u(w, \theta, f) \geq V_u$.
- But since clubs maximize profits, they will not want to offer the unemployed a utility greater than V_u, which is the maximum offered by other clubs.
- This determines the fee charged by a club which picks w and θ :

$$rV_u(w, \theta, f) = -f + \theta q(\theta)(V_e(w) - V_u(w, \theta, f)),$$

$$rV_u = -f + \theta q(\theta)(V_e(w) - V_u(w, \theta, f)),$$

$$f(w, \theta) = \theta q(\theta) V_e(w) - (r + \theta q(\theta)) V_u. \tag{2}$$

2. Search Clubs

Equilibrium (continued)

- Value of posting a vacancy at a club with (w, θ, g) is $V_v(w, \theta, g)$ such that

$$rV_v(w, \theta, g) = -c - g + q(\theta)(J(w) - V_v(w, \theta, g)),$$

- where $J(w)$ is the value to the firm of a position with wage w, i.e.

$$rJ(w) = y - w - sJ(w).$$

- In equilibrium it must be that $V_v(w, \theta, g) \geq 0$
- Profit maximization by clubs will imply $V_v(w, \theta, g) = 0$,
- Therefore $V_v(w, \theta, g) = 0$, implying

$$g(w, \theta) = -c + q(\theta)J(w). \tag{3}$$

2. Search Clubs

Equilibrium (continued)

- Let's look at competition between clubs
- The profit of a club, per unemployed member, is given by $\pi(w, \theta) = f(w, \theta) + \theta g(w, \theta)$.
- Free entry of clubs imply that
  - This cannot exceed zero in equilibrium, for *any* potential club offering w and θ. Therefore :

$$f(w, \theta) + \theta g(w, \theta) \leq 0, \forall w, \theta.$$

  - Profits must be strictly zero for clubs that are nonempty in equilibrium.

2. Search Clubs

Equilibrium (continued)

- Compute $\pi(w, \theta)$ using (2) and (3)

$$\begin{aligned} f(w,\theta) &= \theta q(\theta) V_e(w) - (r + \theta q(\theta)) V_u && (2) \\ g(w,\theta) &= -c + q(\theta) J(w) && (3) \end{aligned}$$

- to get

$$\pi(w,\theta) = \theta q(\theta)(V_e(w) + J(w) - V_u) - (c\theta + rV_u).$$

  - $\theta q(\theta)(V_e(w) + J(w) - V_u)$: benefit of the club, per unemployed member, in terms of job creation.
  - $(c\theta + rV_u)$: cost of search per unemployed member.
- $V_e(w) + J(w) - V_u =$ Total net surplus of a match
- $rV_e(w) = w + s(V_u - V_e(w))$ implies $V_e = \frac{w+sV_u}{r+s}$
- $rJ(w) = y - w - sJ(w)$ implies $J = \frac{y-w}{r+s}$
- Therefore, $V_e(w) + J(w) - V_u = \frac{y-rV_u}{r+s}$

2. Search Clubs

Equilibrium (continued)

- Since $V_e(w) + J(w) - V_u = \frac{y - rV_u}{r+s}$, $\pi(w, \theta) = \pi(\theta)$ does not depend on w.
  - This is due to the fact that w is a transfer between the firm and the worker, which leaves the total surplus of the match unchanged.
  - Any increase in w would be matched by an increase in f and a fall in g to maintain V_u and V_v unchanged.

2. Search Clubs

Equilibrium (continued)

- The preceding equations imply that the equilibrium value of θ, θ^{cs} satisfies

$$\theta^{cs} = \arg\max \pi(\theta) = \theta q(\theta)\frac{y - rV_u}{r+s} - c\theta - rV_u.$$

- Furthermore,

$$\pi(\theta^{cs}) = 0.$$

- These two equations jointly determine θ_E and V_u in equilibrium.
- Note : There is only one type of club in equilibrium.
- Working through those equations yields the equilibrium θ^{cs}, which must satisfy

$$y(q(\theta^{cs}) + \theta^{cs} q'(\theta^{cs})) = c(r + s - (\theta^{cs})^2 q'(\theta^{cs})). \quad (4)$$

3. Comparing equilibrium and optimal value of θ

θ^{cs} and θ^{so}

- Competitive Search Equilibrium θ^{cs} is given by

$$y(q(\theta^{cs}) + \theta^{cs} q'(\theta^{cs})) = c(r + s - (\theta^{cs})^2 q'(\theta^{cs})). \quad (4)$$

- Socially optimal one θ^{so} ?
- Recall from Lecture 10 that Social Planner FOC are

$$c = \lambda_t \frac{\partial m(u, v)}{\partial v} \quad (5)$$

$$r\lambda_t = y - s\lambda_t - \lambda_t \frac{\partial m}{\partial u} + \dot{\lambda}_t \quad (6)$$

3. Comparing equilibrium and optimal value of θ

Computing θ^{so}

- We can re-express those conditions in terms of θ.
- Note that
$$m(u,v) = u\theta q(\theta).$$
- Therefore,
$$\begin{aligned}\frac{\partial m(u,v)}{\partial v} &= u \times [q(\theta) + \theta q'(\theta)]\frac{\partial \theta}{\partial v} \\ &= q(\theta) + \theta q'(\theta).\end{aligned}$$
- Hence
$$\lambda_t = \frac{c}{q(\theta_t) + \theta_t q'(\theta_t)}. \tag{7}$$
- Also,
$$\begin{aligned}\frac{\partial m(u,v)}{\partial u} &= \theta q(\theta) + u[q(\theta) + \theta q'(\theta)]\frac{\partial \theta}{\partial u} \\ &= \theta q(\theta) - \theta[q(\theta) + \theta q'(\theta)] \\ &= -\theta^2 q'(\theta).\end{aligned} \tag{8}$$

3. Comparing equilibrium and optimal value of θ

Computing θ^{so} (continued)

- Substituting (7) and (8) into (6) we get the evolution of θ^{so} :

$$\begin{aligned}\frac{rc}{q(\theta)+\theta q'(\theta)} &= y - \frac{sc}{q(\theta)+\theta q'(\theta)} + \frac{c\theta^2 q'(\theta)}{q(\theta)+\theta q'(\theta)} \\ &+ \frac{d}{dt}\left[\frac{c}{q(\theta_t)+\theta_t q'(\theta_t)}\right].\end{aligned}$$

- This equation only involves θ and is unstable. Therefore the only non explosive solution is the stationary one, i.e.

$$(r+s-\theta^2 q'(\theta))c = \left(q(\theta)+\theta q'(\theta)\right) y. \tag{9}$$

- This determines the optimal $\theta = \theta^{so}$, which is constant over time, like the equilibrium one.

3. Comparing equilibrium and optimal value of θ

$\theta^{cs} = \theta^{so}$

- Competitive search :

$$y(q(\theta^{cs}) + \theta^{cs} q'(\theta^{cs})) = c(r + s - (\theta^{cs})^2 q'(\theta^{cs})). \quad (4)$$

- Optimum :

$$y(q(\theta^{so}) + \theta^{so} q'(\theta^{so})) = c(r + s - (\theta^{so})^2 q'(\theta^{so})). \quad (9)$$